

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator configures a system with IP address **192.168.10.65**, subnet mask **255.255.255.192**, and default gateway **192.168.10.1**, while other systems are configured within the subnet **192.168.10.0/26**. Using the given subnet mask, apply subnetting techniques to determine the network address, broadcast address, and valid host range of the subnet. Verify whether the configured IP address belongs to the given subnet range. Based on your calculations, identify any configuration error related to IP addressing or gateway. Determine the correct IP address and valid gateway that should be assigned to the system. Also, calculate the number of usable host addresses available in the subnet after correction. In your solution, ensure clarity, correctness of calculations, proper step-by-step representation, and accuracy in the final configuration (Code Quality).

2. A company is using the IP block **192.168.1.0/24** and has divided it into subnets using the subnet masks **/26, /27, and /28** for different departments. However, some departments are experiencing IP address shortages while others have unused addresses. Using the given subnet masks, apply subnetting techniques to determine the network address, broadcast address, and valid host range for each subnet. Calculate the number of usable and unused IP addresses in each subnet. Based on your calculations, identify any inefficiencies or mismatches in the current subnet allocation. Suggest corrected or optimized subnet assignments to ensure better utilization of IP addresses within the given network block.

3. A system is configured with the IPv6 address **2001:db8::ff00:42:8329/64**, and another device is configured with **2001:db8:0:0:1::1/64**, but the devices are

unable to communicate. Using IPv6 addressing rules, apply expansion to convert both addresses into their full notation. Determine the network prefix of each address using the given /64 prefix length. Verify whether both addresses belong to the same network prefix. Based on your calculations, identify any addressing or prefix mismatch causing the issue. Suggest the correct IPv6 configuration required to ensure proper communication and calculate the total number of possible host addresses available within the network.

4. A router is configured with the following routing table entries: **192.168.1.0/24** → **next hop 10.0.0.2**, **192.168.2.0/24** → **next hop 10.0.0.3**, and a default route → **10.0.0.1**, but some packets are not reaching the correct destination. A packet arrives with the destination IP address **192.168.2.100**. Using the given routing table, apply routing techniques to identify the matching network for the destination IP address and determine the next-hop address used to forward the packet. Verify whether the destination IP falls within the subnet **192.168.2.0/24**. Based on your findings, identify any possible routing misconfiguration. Suggest the correct or optimized routing entry to ensure proper packet delivery, and also identify the route that will be used if no matching network entry is found.

5. Two LANs are connected using a router, where LAN1 uses the subnet **192.168.5.0/27** and LAN2 uses **192.168.5.32/27**, but communication between the networks is not functioning properly. A PC in LAN1 is assigned the IP address **192.168.5.10**, and a PC in LAN2 is assigned **192.168.5.40**. Using the given subnet information, apply subnetting techniques to determine the network address, broadcast address, and valid host range for each LAN. Verify whether the given IP addresses belong to their respective subnet ranges. Based on your calculations, identify any possible configuration issues related to addressing or gateway settings. Suggest appropriate gateway IP addresses and optimized configuration changes to ensure successful communication between the two

LANs, and calculate the number of usable host addresses available in each subnet.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

